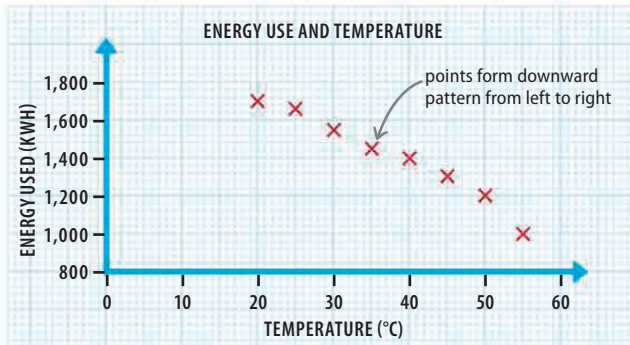


Negative and zero correlations

The points in a scatter diagram can form many different patterns, which reveal different types of correlation between the sets of data. This can be positive, negative, or nonexistent. The pattern can also reveal how strong or how weak the correlation is between the two sets of data.

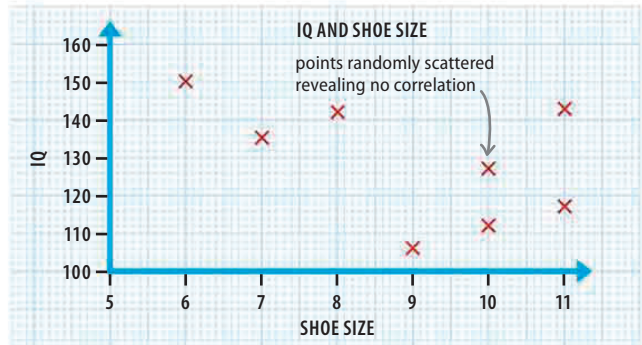
Energy used (kwh)	1,000	1,200	1,300	1,400	1,450	1,550	1,650	1,700
Temperature (°C)	55	50	45	40	35	30	25	20



△ Negative correlation

In this graph, the points form a downward pattern from left to right. This reveals a connection between the two sets of data—as the temperature increases, energy consumption goes down. This relationship is called negative correlation.

IQ	141	127	117	150	143	111	106	135
Shoe size	8	10	11	6	11	10	9	7

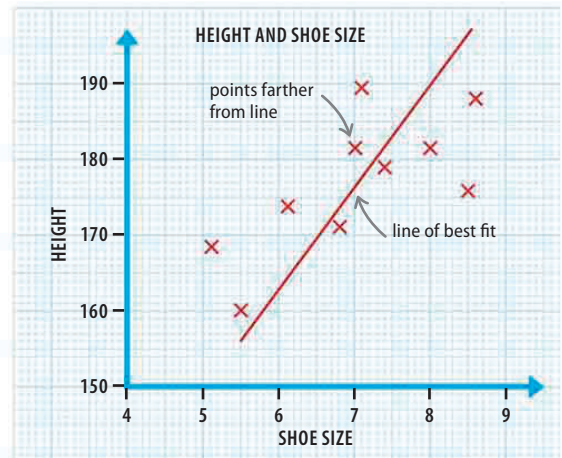
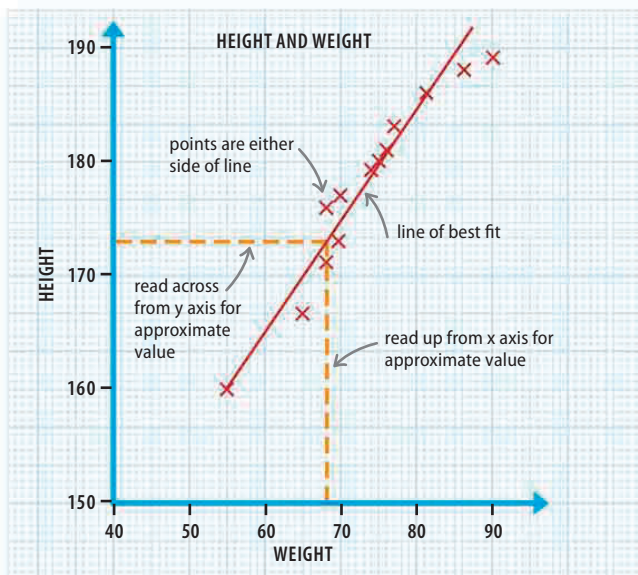


△ No correlation

In this graph, the points form no pattern at all—they are widely spaced and do not reveal any trend. This shows there is no connection between a person's shoe size and their IQ, which means there is zero correlation between the two sets of data.

Line of best fit

To make a scatter diagram clearer and easier to read, a straight line can be drawn that follows the general pattern of the points, with an equal number of points on both sides of the line. This line is called the line of best fit.



△ Weak correlation

Here the points are farther away from the line of best fit. This shows that the correlation between height and shoe size is weak. The farther the points are from the line, the weaker the correlation.

◁ Finding approximate values

When the line of best fit is drawn, approximate values of any weight and height can be found by reading across from the y axis, or up from the x axis.